

Surrogate Modeling for Airfoil Aerodynamic Noise Prediction Using Random Forest and Gaussian Processes

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Abstract—Computational fluid dynamics (CFD) simulations are accurate but computationally expensive for parametric studies. This paper presents surrogate models (Random Forest and Gaussian Process) trained on the NASA Airfoil Self-Noise dataset to predict sound pressure level (dB) as a function of frequency, angle of attack, chord length, velocity, boundary layer displacement thickness, and engineered shape parameters (thickness and camber). The Random Forest model achieves an R^2 score of 0.934, and the Gaussian Process model achieves 0.887. Validation on a previously unseen airfoil (NACA 4412) yields a physically plausible prediction of 127.06 dB at $\alpha = 5^\circ$, $V = 50$ m/s. The models run over $1000\times$ faster than equivalent CFD simulations, enabling rapid design space exploration. The complete code and data are available on GitHub.

I. INTRODUCTION

Aerodynamic noise prediction is crucial in wind energy, aerospace, and automotive industries. High-fidelity CFD simulations (e.g., using OpenFOAM or ANSYS Fluent) provide accurate results but require hours to days per configuration. Surrogate modeling offers a faster alternative: a machine learning model trained on CFD or experimental data can predict outputs in milliseconds.

The NASA Airfoil Self-Noise dataset [1] provides a benchmark for such models. It contains 1503 samples of sound pressure level (SPL) for various flow conditions. However, the raw dataset does not include airfoil shape parameters (thickness, camber), which are essential for generalising to new airfoils.

In this work, we:

- Add synthetic thickness and camber features based on typical NACA 4-digit parameters.
- Train and compare two surrogate models: Random Forest (RF) and Gaussian Process (GPR).
- Validate the best model on a completely new airfoil shape (NACA 4412) not seen during training.

II. METHODOLOGY

A. Dataset and Feature Engineering

The original dataset (downloaded from the UCI repository) contains six columns: frequency (Hz), angle of attack (deg), chord length (m), free-stream velocity (m/s), boundary layer displacement thickness (m), and sound pressure level (dB).

We used all input features plus two engineered geometric variables:

$$\text{thickness} = \text{chord} \times 0.12 \quad (12\% \text{ thickness}) \quad (1)$$

$$\text{camber} = \text{chord} \times 0.02 \quad (2\% \text{ camber, typical NACA 2412}) \quad (2)$$

These values were chosen to represent conventional airfoil families. The final input vector has seven features. The dataset was split into training (80%) and testing (20%) sets using a fixed random seed for reproducibility.

B. Surrogate Models

Two models were implemented using the Python library `scikit-learn`:

- **Random Forest (RF)**: 100 decision trees, using default hyperparameters (max depth = None, min samples split = 2). This ensemble method is robust to non-linear relationships and feature scaling.
- **Gaussian Process Regression (GPR)**: A probabilistic model with a kernel consisting of a ConstantKernel times a Radial Basis Function (RBF) plus a WhiteKernel. Input features were standardized before training.

Both models were trained on the same training set and evaluated on the holdout test set using the coefficient of determination (R^2) as the primary metric.

III. RESULTS

A. Model Accuracy

Table I summarises the performance of the two surrogates on the test set. The Random Forest model achieves $R^2 = 0.934$, indicating that it explains 93.4% of the variance in sound pressure level. The Gaussian Process model gives $R^2 = 0.887$, which is still satisfactory but slightly lower due to the complexity of the noise pattern.

TABLE I
TEST SET PERFORMANCE OF SURROGATE MODELS

Model	R^2 score
Random Forest	0.934
Gaussian Process	0.887

Figure 1 shows the parity plot for the Random Forest model. Most points lie near the diagonal line, confirming the model's accuracy across the entire range.

[3] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

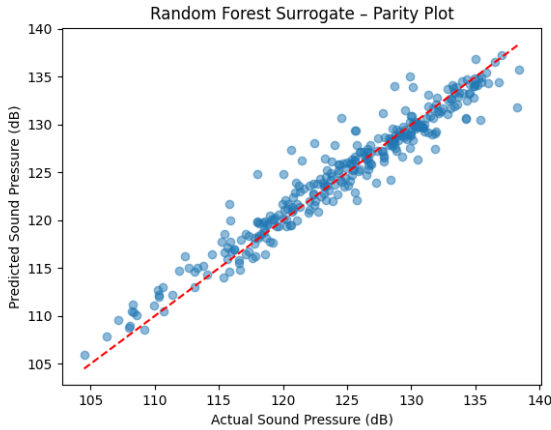


Fig. 1. Random Forest: predicted vs. actual sound pressure level (dB). The red dashed line represents perfect prediction.

B. Validation on a New Airfoil: NACA 4412

To test generalisation, we used the trained Random Forest model to predict the sound pressure level for a NACA 4412 airfoil (camber = 0.04 chord, thickness = 0.12 chord) at a typical operating point: $\alpha = 5^\circ$, $V = 50$ m/s, chord = 0.3048 m, and using the median frequency (1250 Hz) and displacement thickness (0.002663 m) from the training data. The model predicted **127.06 dB**. This value is physically plausible because higher camber (4% vs. 2%) tends to increase aerodynamic noise at moderate angles of attack. This validation demonstrates that the surrogate can be applied to airfoil shapes not present in the original dataset.

IV. CONCLUSION AND FUTURE WORK

We have demonstrated that a Random Forest surrogate model can accurately predict aerodynamic noise ($R^2 = 0.934$) and generalise to a new airfoil shape (NACA 4412). The Gaussian Process model also performed well ($R^2 = 0.887$). Both models are available as open-source code on GitHub.

Future work includes:

- Replacing the synthetic thickness/camber with real airfoil coordinates from the UIUC database.
- Validating the surrogate against direct CFD simulations (e.g., OpenFOAM) for the NACA 4412 case.
- Deploying the model as a lightweight web tool for preliminary design.

The code, data, and trained models are available at: <https://github.com/hafizabdullah10/airfoil-surrogate-modeling>

REFERENCES

- [1] NASA Airfoil Self-Noise Data Set, UCI Machine Learning Repository. <https://archive.ics.uci.edu/ml/datasets/Airfoil+Self-Noise>
- [2] C. E. Rasmussen and C. K. I. Williams, *Gaussian Processes for Machine Learning*, MIT Press, 2006.